Longlong Li

PhD Student | BioMEMS / Biosensors / Engineered Cardiac Tissue Monitoring

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Comprehensive academic version for research collaboration, postdoctoral applications, and academic review.

Profile

- PhD student specializing in BioMEMS, multimodal biosensors, engineered cardiac tissue monitoring, and drug cardiotoxicity evaluation. Experienced in MEMS device design, microfabrication, cell and tissue culture, sensing system integration, and multimodal biosignal analysis. My research focuses on integrating strain sensors, microelectrode arrays, and 3D bioelectronic platforms for synchronized and quantitative assessment of cardiac tissue function.

Education

- 2022-Present | Ph.D. in Mechanical Engineering, Chonnam National University, Gwangju, South Korea
- 2018-2022 | B.Eng. in Mechanical Engineering, Wenzhou University, Wenzhou, China

Research Experience

- Research experience 1 | Designed and developed BioMEMS-based multimodal sensing platforms for engineered cardiac tissues.
- Research experience 2 | Integrated strain sensors and microelectrode arrays to monitor mechanical contraction and electrical activity in cardiac tissue models.
- Research experience 3 | Established 3D engineered heart tissue culture and drug stimulation workflows for cardiotoxicity screening and disease modeling.
- Research experience 4 | Developed MATLAB / Python-based analysis workflows for extracting contraction amplitude, beating frequency, field potential features, and electromechanical coupling parameters.
- Research experience 5 | Built experimental measurement systems involving low-noise signal acquisition, microscopy, sensor calibration, and hardware integration.

Technical Skills

- Microfabrication | Photolithography, metal deposition, SU-8 / PI processing, MEMS sensor fabrication, device packaging, sensor characterization.
- Biosensing | Strain sensors, microelectrode arrays, electromechanical monitoring, low-noise signal acquisition, sensor calibration.
- Cell & Tissue Engineering | NRVM / hiPSC-CMs culture, engineered heart tissue construction, collagen / dECM bioinks, drug stimulation experiments, immunostaining.
- Data Analysis | MATLAB, Python, cardiac signal processing, feature extraction, visualization, statistical analysis.
- System Integration | DAQ systems, amplifier circuits, microscopy, automated measurement platforms, prototype development.

Projects

- 2026.03-Present | Bioinspired Hypoxic Cardiac Model and Electromechanical Evaluation Platform | MNTL | Developed an in vitro cardiac hypoxia model combined with multimodal electromechanical readouts. | Integrated mechanical contraction sensing and electrical activity recording for functional assessment of engineered cardiac tissues. | Applied the platform to evaluate cardiac tissue response under disease-relevant microenvironmental conditions.
- 2024.03-2026.03 | High-Throughput Drug Toxicity Screening Platform Based on 3D Cardiac Tissues | MNTL | Designed and optimized a 3D cardiac tissue-based sensing platform for drug cardiotoxicity evaluation. | Integrated strain-based mechanical sensing and electrophysiological readouts. | Supported long-term monitoring and multiparametric analysis of drug-induced functional responses.

- 2022.09-2024.03 | Graphene-Conductive Microenvironment for Cardiomyocyte Maturation and Electrophysiological Communication | MNTL | Investigated conductive microenvironment effects on cardiomyocyte maturation and intercellular communication. | Developed graphene/SU-8-based platforms for cardiac tissue functional enhancement. | Contributed to journal publication and conference outputs.

Publications

- Air-Breakdown Triboelectric Nanogenerator Inspired by Transistor Architecture for Low-Force Human-Machine Interfaces. Karthikeyan Munirathinam, Longlong Li, Arunkumar Shanmugasundaram, Jongsung Park, Dong-Weon Lee. Nano-Micro Letters, 18(1), 251, 2026. Link: 10.1007/s40820-026-02103-0

- Tilted-angle acoustofluidic separation of live and dead neonatal rat ventricular myocytes using hypotonic cell swelling. Mushtaq Ali, Qu Zhuji, Nomin-Erdene Oyunbaatar, Li Longlong, Dong-Weon Lee, Jinsoo Park. Sensors and Actuators B: Chemical, 2026. Link: https://www.sciencedirect.com/science/article/pii/S0925400526006970

- Alginate-derived smart hydrogel dressing with CaP/ZnO synergy for on-demand hemostasis and pH-triggered curcumin delivery. Kamrun Nahar Fatema, Md Rokon Ud Dowla Biswas, Longlong Li, Biswajit Mahanty, Khurshid Ahmad, Ishrat Jahan, Sungtae Yang, Jongyun Kim, Dong-Weon Lee. Journal of Drug Delivery Science and Technology, 125, 108658, 2026. Link: 10.1016/j.jddst.2026.108658

- Strain Sensor–Integrated Tri-Layer Cantilever with Enhanced Sensitivity and Stability for Quantitative Cardiotoxicity Screening. Ke Liu, Arunkumar Shanmugasundaram, Haolan Sun, Longlong Li, Gowoon Jeong, Seong Young Ko, Eung-Sam Kim, Bong-Kee Lee, Dong-Weon Lee. ACS Sensors, 2026. Link: 10.1021/acssensors.6c01331

- Integrated Cantilever–MEA Platform for High-Sensitivity and Scalable Recording of Cardiomyocyte Electromechanics. Haolan Sun, Longlong Li, Arunkumar Shanmugasundaram, Eung-Sam Kim, Bong-Kee Lee, Dong-Weon Lee. Small, e74949, 2026. Link: 10.1002/smll.74949

- A Self-Coupling Microcantilever-Probe Platform: Non-compressive Sustained Assessment of Cardiac Contractility and Cardiotoxicity Screening. Haolan Sun, Longlong Li, Eung-Sam Kim, Bong-Kee Lee, Dong-Weon Lee. Sensors and Actuators Reports, 100516, 2026. Link: 10.1016/j.snr.2026.100516

- Harnessing native blueprints for designing bioinks to bioprint functional cardiac tissue. Mst Zobaida Akter, Fatima Tufail, Ashfaq Ahmad, Yoon Wha Oh, Jung Min Kim, Seoyeon Kim, Md Mehedee Hasan, Longlong Li, Dong-Weon Lee, Yong Sook Kim, Su-jin Lee, Hyung-Seok Kim, Youngkeun Ahn, Yeong-Jin Choi, Hee-Gyeong Yi. iScience, 28(3), 2025. Link: https://www.cell.com/iscience/fulltext/S2589-0042(25)00142-7

- Development of multifunctional PAA-alginate-carboxymethyl cellulose hydrogel-loaded fiber-reinforced biomimetic scaffolds for controlled release of curcumin. Kamrun Nahar Fatema, Longlong Li, Khurshid Ahmad, Jongyun Kim, Dong-Weon Lee. International Journal of Biological Macromolecules, 301, 140449, 2025. Link: 10.1016/j.ijbiomac.2025.140449

- Dual-sensitized hollow SnO2 nanospheres with rGO and Pd for highly sensitive detection of acetone in exhaled breath. Arunkumar Shanmugasundaram, Kun Woo Baek, Changung Paeng, Longlong Li, Goeun Cha, Jonghyeon Woo, Dong-Su Kim, Changyong Yim, Jongsung Park, Jung Sang Cho, Dong-Weon Lee. Applied Surface Science, 696, 162959, 2025. Link: https://www.sciencedirect.com/science/article/pii/S0169433225006737

- Hydrogel-Integrated Biomimetic Hydroxyapatite Scaffolds with Tunable Porosity for Enhanced Curcumin Delivery. Kamrun Nahar Fatema, Longlong Li, Khurshid Ahmad, Jongyun Kim, Dong-Weon Lee. Journal of Drug Delivery Science and Technology, 107572, 2025. Link: https://www.sciencedirect.com/science/article/pii/S177322472500975X

- Enhancing cardiomyocyte maturation through PEDOT:PSS-coated surfaces and mechanical stimulation with strain sensors. Jongyun Kim, Arunkumar Shanmugasundaram, Longlong Li, Zhuo Qu, Eung-Sam Kim, Bong-Kee Lee, Dong-Weon Lee. Journal of Micromechanics and Microengineering, 35(4), 045002, 2025. Link: 10.1088/1361-6439/adb8f9

- InGaN-GaN-MQW-ZnO based e-nose sensors for nitrogen dioxide detection using advanced machine learning approaches. Arunkumar Shanmugasundaram, Mandar A. Kulkarni, Changung Paeng, Jonghyeon Woo, Heonzoo Lee, Longlong Li, Huijin Lee, Changyong Yim, Yonggwan Won, Sang-Wan Ryu, Jongsung Park, Dong-Weon Lee. Sensors and Actuators B: Chemical, 138650, 2025. Link: https://www.sciencedirect.com/science/article/pii/S0925400525014261

- Graphene SU-8 platform for enhanced cardiomyocyte maturation and intercellular communication in cardiac drug screening. Longlong Li, Arunkumar Shanmugasundaram, Jongyun Kim, Nomin-Erdene Oyunbaatar, Pooja P. Kanade, Seong-Eung Cha, Daeyun Lim, Chil-Hyoung Lee, Young-Baek Kim, Bong-Kee Lee, Eung-Sam Kim, Dong-Weon Lee. ACS Nano, 18(49), 33293-33309, 2024. Link: 10.1021/acsnano.4c05365

Conferences - Oral Presentations

- Real-Time Electromechanical Monitoring of Engineered Heart Tissue Responses to Cardiac Drugs. Longlong Li, Haolan Sun, Jong-Yun Kim, Dong-Weon Lee. KSPE 2026, Jeju, Korea, 2026.05.13

- Multimodal Microelectrode-Microcantilever Array for Electromechanical Analysis of Cardiomyocyte Tissue in Drug Testing. Haolan Sun, Longlong Li, Eung-Sam Kim, Bong-Kee Lee, Dong-Weon Lee. IEEE MEMS 2026, Salzburg, Austria, 2026.01.26. Link: https://ieeexplore.ieee.org/abstract/document/11419567

- 3D Cardiac Tissue Platform for Monitoring Contractility and Assessing Cardiotoxicity. Longlong Li, Haolan Sun, Jong-Yun Kim, Dong-Weon Lee. KMEMS 2025, Jeju, Korea, 2025.05.30

- A Dual-Detection Approach for Cardiotoxicity Screening: Utilizing Nano Silicon Strain Sensor and Mea to Monitor Contractility and Field Potential in Cardiomyocytes. Haolan Sun, Longlong Li, Dong-Weon Lee. IEEE MEMS 2025, Kaohsiung, Taiwan, 2025.01.20. Link: https://ieeexplore.ieee.org/abstract/document/10917809

- Enhancing Cardiomyocyte Maturation via Mechanical Stimulation of 3D Printed Cardiac Tissue Using a Origami-based 3D Sensor and Magnetic Fields. Longlong Li, Haolan Sun, Jong-Yun Kim, Yun-Jin Jeong, Bong-Kee Lee, Eung-Sam Kim, Dong-Weon Lee. IEEE NANOMED 2024, Hawaii, USA, 2024.12.04. Link: https://ieeexplore.ieee.org/abstract/document/10946052

- The hybrid cantilever of conductive graphene and su-8 for improving the maturity and electrical coupling of cardiomyocytes. Longlong Li, Jong-Yun Kim, Daeyun Lim, Chil-Hyoung Lee, and Dong-Weon Lee. MicroTAS 2023, Katowice, Poland, 2023.10.18

Conferences - Poster Presentations

- A Flexible Electromechanical Sensing Platform for In Vitro Hypoxic Cardiac Tissue Models. Longlong Li, Haolan Sun, Jongyun Kim, Dong-Weon Lee. MicroTAS 2026, Granada, Spain, 2026.10.21

- Backside DRIE Release of Grooved SU-8 Cantilever Arrays for Low Warpage and Reproducible Cardiomyocyte Beating Readout. Xin Luo, Hong-Mo Koo, Haolan Sun, Longlong Li, Dong-Weon Lee. MicroTAS 2026, Granada, Spain, 2026.10.20

- Enhancement of cardiomyocyte microenvironment and functional assessment using through-hole structures with integrated polymer cantilevers. Fusong Li, Haolan Sun, Longlong Li, Jong-Yun Kim, Dong-Weon Lee. MicroTAS 2025, Adelaide, Australia, 2025.11.04

- Development of a Multi-Channel Wireless Monitoring Platform for Long-Term Cardiomyocyte Contraction Assessment Using a Polymer Cantilever with integrated Sensor. Ke Liu, Haolan Sun, Longlong Li, Dong-Weon Lee. MicroTAS 2025, Adelaide, Australia, 2025.11.03

- Origami-Inspired 3D Sensor Platform for Real-Time Electromechanical Coupling Analysis in Engineered Heart Tissues. Longlong Li, Fatima Tufail, Haolan Sun, Jongyun Kim, Hee-Gyeong Yi, Dong-Weon Lee. IEEE SENSORS 2025, Vancouver, Canada, 2025.10.20. Link: https://ieeexplore.ieee.org/abstract/document/11330375

- Integrated Bioelectronic Platform Utilizing PEDOT: PSS Strain Sensors for Real-Time Mechanostimulation and Sensing. Zhuji Qu, Jongyun Kim, Haolan Sun, Longlong Li, Dong-Weon Lee. IEEE SENSORS 2025, Vancouver, Canada, 2025.10.20. Link: https://ieeexplore.ieee.org/abstract/document/11331195

- MONITORING OF DRUG-IMFACTED CARDIOMYOCYTES CONTRACTILITY USING PI MICROCANTILEVER STRUCTURES WITH NANO-SILICON STRAIN SENSOR. Haolan Sun, Longlong Li, Dong-Weon Lee. MicroTAS 2024, Montreal, Canada, 2024.10.15

- Polymer Cantilever Integrated with a Full-Bridge Sensor for Continuous Wireless Measurement of Cardiomyocyte Contractility. Ke Liu, Haolan Sun, Longlong Li, Dong-Weon Lee. MicroTAS 2024, Montreal, Canada, 2024.10.14

Patents

- 15 total patents | 3 invention patents | 3 granted inventions | 11 utility model patents

- Vertical-Folding-Based Clothes Folding Machine. Inventors: Wu Mingge, Wei Jinliang, Li Longlong, Yang Yang, Huang Andong, Bao Haibo, Shen Yunde. Patentee / Applicant: Wenzhou University. Patent No. / Application No.: CN113186699B / 202110624021.0. Status: Granted invention patent. Application date: 2021.06.04. Grant / publication date: 2023.02.07

- Automatic Potato Harvester. Inventors: Li Longlong, Shen Yunde, Yang Yang, Wang Tao, Wu Mingge. Patentee / Applicant: Wenzhou University. Patent No. / Application No.: CN216415123U / 202122859852.4. Status: Utility model

patent. Application date: 2021.11.22. Grant / publication date: 2022.05.03

- Potato Digging Device. Inventors: Yang Yang, Wu Mingge, Li Longlong, Yang Demin, Shen Yunde. Patentee / Applicant: Wenzhou University. Patent No. / Application No.: CN216313996U / 202122864112.X. Status: Utility model patent. Application date: 2021.11.22. Grant / publication date: 2022.04.19

- Intelligent Launch-and-Recovery Management Device for UAV Swarms. Inventors: Yang Yang, Zhou Chen, Li Longlong, Chen Haodong. Patentee / Applicant: Wenzhou University. Patent No. / Application No.: CN216269980U / 202123168608.X. Status: Utility model patent. Application date: 2021.12.16. Grant / publication date: 2022.04.12

- Potato-Soil Separation Device for Potato Harvesters. Inventors: Li Longlong, Ma Guang, Yang Yang, Wang Tao, Wu Mingge, Shen Yunde. Patentee / Applicant: Wenzhou University. Patent No. / Application No.: CN216253977U / 202122864125.7. Status: Utility model patent. Application date: 2021.11.22. Grant / publication date: 2022.04.12

- Traction Device for Potato Harvesters. Inventors: Yang Yang, Ma Guang, Li Longlong, Yin Zhi, Wu Mingge, Shen Yunde. Patentee / Applicant: Wenzhou University. Patent No. / Application No.: CN216232634U / 202122864096.4. Status: Utility model patent. Application date: 2021.11.22. Grant / publication date: 2022.04.08

- Trash Bin Based on Cam-Track Bag Closing and Heat Sealing. Inventors: Yang Yang, Han Yifei, Li Longlong, Pan Weijie, Liu Jintao, Shen Yunde. Patentee / Applicant: Cixi Zhuoshang Electric Appliance Co., Ltd.; Wenzhou University. Patent No. / Application No.: CN216234167U / 202122813075.X. Status: Utility model patent. Application date: 2021.11.17. Grant / publication date: 2022.04.08

- Household Intelligent Toy Storage Robot. Inventors: Li Longlong, Zhou Chen, Gao Kewei, Wu Yu, Chen Haodong, Zhang Zhiquan. Patentee / Applicant: Wenzhou University. Patent No. / Application No.: CN112407980B / 202011170715.3. Status: Granted invention patent. Application date: 2020.10.28. Grant / publication date: 2022.01.18

- Automatic Potato Harvester. Inventors: Shen Yunde, Li Longlong, Yang Yang, Wu Mingge, Zhang Chenghao. Patentee / Applicant: Wenzhou University. Patent No. / Application No.: CN113875385A / 202111383414.3. Status: Published invention patent application. Application date: 2021.11.22. Grant / publication date: 2022.01.04

- Telescopic Rotating Arm for Folding and Rolling in a Clothes Folding Machine. Inventors: Li Longlong, Wu Mingge, Bao Haibo, Shen Yunde. Patentee / Applicant: Wenzhou University. Patent No. / Application No.: CN215163984U / 202121243359.3. Status: Utility model patent. Application date: 2021.06.04. Grant / publication date: 2021.12.14

- Zinc Alloy Dross-Skimming Robot and Working Method. Inventors: Ma Guang, Sun Haolan, Wei Jinliang, Chen Haodong, Li Longlong, Gao Kewei, Shen Yunde. Patentee / Applicant: Wenzhou University. Patent No. / Application No.: CN111923063B / 202010801270.8. Status: Granted invention patent. Application date: 2020.08.11. Grant / publication date: 2021.07.16

- Timing-Belt Toy Collection Mechanism for a Household Intelligent Toy Storage Robot. Inventors: Gao Kewei, Cheng Taihong, Li Longlong, Chen Haodong, Zhang Zhiquan, Wu Yu. Patentee / Applicant: Wenzhou University. Patent No. / Application No.: CN213536204U / 202022431641.6. Status: Utility model patent. Application date: 2020.10.28. Grant / publication date: 2021.06.25

- Cart-Bucket Turning Mechanism for Dumping Objects. Inventors: Zhang Zhiquan, Jiang Rui, Wu Yu, Chen Haodong, Gao Kewei, Li Longlong. Patentee / Applicant: Wenzhou University. Patent No. / Application No.: CN213443970U / 202022431595.X. Status: Utility model patent. Application date: 2020.10.28. Grant / publication date: 2021.06.15

- Synchronous Multi-Scissor Lifting Device for Object Lifting. Inventors: Wu Yu, Zhou Chen, Chen Haodong, Zhang Zhiquan, Gao Kewei, Li Longlong. Patentee / Applicant: Wenzhou University. Patent No. / Application No.: CN213446006U / 202022431602.6. Status: Utility model patent. Application date: 2020.10.28. Grant / publication date: 2021.06.15

- Zinc Alloy Dross-Skimming Robot. Inventors: Sun Haolan, Ma Guang, Wei Jinliang, Chen Haodong, Li Longlong, Gao Kewei, Shen Yunde. Patentee / Applicant: Wenzhou University. Patent No. / Application No.: CN212763486U / 202021656863.1. Status: Utility model patent. Application date: 2020.08.11. Grant / publication date: 2021.03.23

Awards

- 2026.09 | Outstanding Paper Award, KMEMS 2026 | Conference paper award | KMEMS 2026 | Co-author of the award-winning poster “Long-Term Spatially Electromechanical Monitoring of hiPSC-CMs Reveals Progressive Doxorubicin Cardiotoxicity” The Society of Micro and Nano Systems 2026

- 2026.09 | Outstanding Paper Award, KMEMS 2026 | Conference paper award | KMEMS 2026 | Co-author of the award-winning poster “A Through-Hole Cantilever Array Integrated with High-Sensitivity Sensors for Functional Monitoring of Bilayer Cardiomyocyte Tissues” The Society of Micro and Nano Systems 2026

- 2026.07 | Best Poster Presentation Award, JCK MEMS/NEMS 2026 | Conference poster award | JCK MEMS/NEMS 2026 | Co-author of the award-winning poster “A PDMS-Based Double-Layer Platform with Integrated Strain Sensing for Controlled Mechanical Conditioning of Cardiomyocytes” Japan–China–Korea Joint Conference on MEMS/NEMS 2026
- 2025.08 | Best Poster Presentation Award, KSS 2025 | Conference poster award | KSS 2025 | Co-author of the award-winning poster “PI Through-Hole Cantilever Array Integrated with Silicon Strain Sensors for Cardiomyocyte Contractility Measurement and High-Throughput Drug Screening” Conference of the Korean Sensors Society 2025
- 2025.03 | Outstanding Paper Award, KMEMS 2025 | Conference paper award | KMEMS 2025 | First author of the award-winning oral presentation “3D Cardiac Tissue Platform for Monitoring Contractility and Assessing Cardiotoxicity” The Society of Micro and Nano Systems 2025
- 2025.02 | RLRC Outstanding Graduate Student | Institute-level honor | RLRC | Academic recognition
- 2025 | BK21 Fellowship Scholarship | BK21 | Scholarship / fellowship support
- 2021 | China National Scholarship | National-level scholarship | China | National-level scholarship, China
- 2021 | 18th Zhejiang Mechanical Design Competition (Kitchen Machinery) | Provincial Second Prize | Provincial Second Prize, participant
- 2020 | 3rd China University Intelligent Robot Creativity Competition | National Second Prize | National Second Prize, participant
- 2020 | 12th National College Student Mathematics Competition | National Third Prize | National Third Prize
- 2020 | 17th Zhejiang Mechanical Design Competition (Smart Home) | Provincial First Prize | Provincial First Prize, team lead
- 2019 | Zhejiang Provincial Government Scholarship | Provincial-level scholarship | Zhejiang Provincial Government | Provincial-level scholarship, Zhejiang, China
- 2019 | 15th Zhejiang University Games (Roller Hockey) | Provincial First Prize | Provincial First Prize, participant
- 2019 | 1st Zhejiang Intelligent Robot Creativity Competition | Provincial Second Prize | Provincial Second Prize, participant